

Training-Free Behavioral Drift Detection in Large Language Models Using Output-Embedding Stability Monitoring

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Abstract

Large language models can exhibit behavioral drift under changes in sampling temperature, exposure to safety-sensitive prompts, fine-tuning, and adversarial prompt injection. Many recent related methods either require access to model internals such as hidden states, attention weights, or token probability distributions, or operate on input prompt embeddings rather than generated outputs. This paper describes a training-free stability metric computed from generated outputs alone and evaluates it across eight validation tests centered on TinyLlama 1.1B-Chat, with additional validation on Phi-2 2.7B, GPT-2, and a base TinyLlama checkpoint. The metric is defined as $\Phi = I \times \rho - \alpha \times S$, where I is an identity component computed from cosine similarity between baseline and comparison output embeddings, ρ is a temporal coherence component measured across sequential evaluation windows, S is an output variation component derived from multiple sampled outputs, and α is a coupling constant fixed at 0.1. The tested settings produced measurable drops in Φ across four tested drift conditions: a 0.282 drop under temperature induced quality degradation on TinyLlama, a 0.072 drop accompanied by a refusal rate collapse from 14% to 3% on safety sensitive prompts, a 0.254 drop between a base and instruction tuned checkpoint pair, and a 0.207 drop under an adversarial prompt injection wrapper with the same seeds in both conditions. The same safety-regime evaluation framework reproduced on Phi-2 with a 0.044 drop using the same temperature pair and prompt suite with a reduced sample count for runtime reasons, and Φ tracked sampling temperature continuously with a correlation of -0.97 across eight temperature points. The method operates in black-box settings using only generated text and an external embedding model. Results are bounded to the tested models, prompt suites, and configurations; broader generalization is not established.

1 Introduction

Large language models deployed in production are subject to multiple forms of behavioral drift. Sampling temperature changes alter output distributions. Safety-sensitive prompts may elicit different behavior under different generation parameters. Fine-tuning operations shift model behavior in ways that are not always measured against the pre-fine-tuning checkpoint. Adversarial prompt injection wrappers can redirect model behavior

without modifying the underlying weights. Each of these conditions produces measurable changes in generated outputs, but the detection methods available for each condition typically differ from the others and often require privileged access to model internals.

This paper examines whether a single training-free stability metric, computed from generated text outputs alone, detects behavioral drift across all four of these conditions. The metric is defined as $\Phi = I \times \rho - \alpha \times S$, where I is an identity component computed from cosine similarity between baseline and comparison output embeddings, ρ is a temporal coherence component measured across sequential evaluation windows, S is an output variation component derived from multiple sampled outputs under fixed configuration, and α is a coupling constant fixed at 0.1. The components are computed using an external embedding model that operates independently of the language model under evaluation.

The paper addresses four questions. First, does Φ detect quality degradation induced by sampling temperature on a small instruction-tuned model? Second, does the same metric detect safety-regime behavioral drift on safety-sensitive prompts, and does the detected drift co-track with refusal rate changes on those prompts? Third, does the metric reproduce on a multi-billion-parameter model with the same formula and the same output-only evaluation framework? Fourth, does the metric detect behavioral shifts between a base model checkpoint and an instruction-tuned checkpoint, and behavioral shifts induced by adversarial prompt injection with matched sampling seeds?

The contributions are as follows. The paper reports eight validation tests centered on TinyLlama 1.1B-Chat, with additional comparisons involving Phi-2 2.7B, GPT-2, and a pre-instruction-tuning TinyLlama checkpoint. The eight tests cover baseline reproducibility, quality drift, safety-regime drift, temporal coherence, temperature sensitivity, discrimination between the tested model architectures and checkpoints, fine-tuning drift, and adversarial injection drift. A separate scale-validation run on Phi-2 2.7B reproduces the safety-regime test with the same formula and a reduced-sample version of the same evaluation framework. All tests use real inference on the specified models with no synthetic generations. Test 1 serves as a baseline reproducibility check rather than a hardened drift test. Tests 3, 6, 7, and 8 are fully hardened with embedding revision pinning, per-output hashes, and manifest SHA256 records. Test 4 is partially hardened with embedding revision pinning and per-output hashes but no manifest. Tests 2 and 5 are not hardened and are carried as supporting evidence. The paper reports both the measured drift signals and the scope bounds that limit their interpretation.

Results are bounded to the tested models, prompt suites, and configurations. Whether the metric generalizes to larger models beyond the 2.7B parameter scale, to out-of-distribution safety categories, to other fine-tuning regimes, or to adversarial attack patterns beyond the tested injection wrapper is not established by the evidence reported here.

2 Related Work

Recent work on language-model drift detection falls into several categories. Some methods compare embeddings of input prompts to detect prompt injection, with the drift signal computed on the input side rather than on generated outputs (1). Other methods monitor token-level entropy or other generation-time statistics, which requires access to token probability distributions exposed by the language model (2). A separate class of methods studies behavioral drift across long interaction traces or multi-agent settings, where drift

accumulates over extended dialogue rather than appearing under configuration changes (3). Another class uses internal model representations to detect factual inconsistencies or truthfulness drift, which requires access to hidden states or other privileged model internals (4).

The present approach differs from these lines of work in three respects. First, embeddings are computed on generated outputs rather than on input prompts. Second, the detection signal is a continuous three-term scalar that combines an identity component, a temporal coherence component, and an output variation component, rather than a binary classification or a single-term signal. Third, the method operates in black-box settings using only generated text and an external embedding model, without access to hidden states, attention weights, or token probability distributions.

A separate line of work studies hallucination and output uncertainty through multi-sample variation, typically by clustering multiple sampled answers by semantic meaning and measuring entropy across the resulting clusters (5). The S component used in the present paper differs in that it is a bounded scalar derived from variation among sampled output embeddings under fixed configuration, used as one term inside the three-term Φ formula rather than as a standalone hallucination signal.

Other work studies behavioral drift diagnostically under semantically equivalent prompt rewordings or other controlled perturbations, with embedding-based metrics demonstrating that drift can be measured across models and temperatures (6). The present paper uses a similar output-embedding approach but differs in producing a single scalar that is reported with comparable interpretation across four distinct drift conditions in the same formulation.

3 Methods

3.1 The Stability Metric

The stability metric Φ is a scalar computed from three components:

$$\Phi = I \times \rho - \alpha \times S \quad (1)$$

Throughout this section, E denotes an external embedding model that maps a text output to a fixed-dimension vector, and all individual output embeddings are L^2 -normalized before use. Given two conditions A and B (for example, a baseline and a comparison temperature), the language model produces outputs $\{o_{ij}^A\}$ and $\{o_{ij}^B\}$ where $i \in \{1, \dots, N\}$ indexes prompts and $j \in \{1, \dots, K\}$ indexes samples per prompt. The normalized sample embeddings are:

$$e_{ij}^A = \frac{E(o_{ij}^A)}{\|E(o_{ij}^A)\|}, \quad e_{ij}^B = \frac{E(o_{ij}^B)}{\|E(o_{ij}^B)\|}. \quad (2)$$

Identity similarity I . The condition-level mean embedding is computed by averaging all NK normalized sample embeddings within a condition and L^2 -normalizing the result:

$$\bar{e}^A = \frac{\mu^A}{\|\mu^A\|}, \quad \mu^A = \frac{1}{NK} \sum_{i=1}^N \sum_{j=1}^K e_{ij}^A, \quad (3)$$

and analogously for $\bar{e}^B$. The identity similarity between the two conditions is the cosine similarity of the two mean embeddings, clamped to $[0, 1]$:

$$I = \text{clip}_{[0,1]}(\bar{e}^A \cdot \bar{e}^B). \quad (4)$$

In the two-condition drift reporting used in this paper, condition A is treated as the reference, so $I_A = 1.0$ by definition and I_B equals I as computed above. The quantity I_B is reported in the results tables as I_{drift} .

Output variation S . For a single condition, let e_{ij} denote the L^2 -normalized embedding of sample j under prompt i . The output variation component is the mean pairwise cosine distance across the K samples per prompt, averaged over the N prompts, divided by 2 to map the cosine-distance range $[0, 2]$ into $[0, 1]$, and clamped to $[0, 1]$:

$$S = \text{clip}_{[0,1]} \left(\frac{1}{2N} \sum_{i=1}^N \frac{2}{K(K-1)} \sum_{1 \leq j < k \leq K} (1 - e_{ij} \cdot e_{ik}) \right). \quad (5)$$

For $K < 2$ (insufficient samples for a pairwise comparison within a prompt), S is defined as 0.

Temporal coherence ρ . For single-snapshot tests comparing two fixed conditions (Tests 2, 3, 3+, 5, 6, 7, and 8), $\rho = 1.0$ by construction, and Φ reduces to $I - \alpha \times S$. For windowed tests (Test 4), the prompt suite is evaluated across W sequential windows, and a mean embedding $\bar{e}_w$ is computed per window using the same mean-and-renormalize formula above. The per-window temporal coherence is the cosine similarity between adjacent-window mean embeddings, clamped to $[0, 1]$:

$$\rho_w = \text{clip}_{[0,1]}(\bar{e}_w \cdot \bar{e}_{w-1}) \quad \text{for } w = 2, \dots, W, \quad (6)$$

and the global ρ reported for a windowed test is the mean of the per-window values:

$$\rho = \frac{1}{W-1} \sum_{w=2}^W \rho_w. \quad (7)$$

Values of ρ close to 1.0 indicate that adjacent windows produce similar mean embeddings; values close to 0.0 indicate substantial behavioral drift across windows.

The coupling constant α is fixed at 0.1 across all tests in this paper. No threshold is used to gate pass or fail at the protocol layer. Φ is logged for each test condition and reported as a continuous measurement. The reference implementation of these definitions is in `core/phi_llm.py` in the accompanying repository.

3.2 The Language Models and the Embedding Model

The language models used in this paper are as follows. TinyLlama 1.1B-Chat-v1.0 (7) is the primary test bed across seven of the eight tests. TinyLlama 1.1B-intermediate-step-1431k-3T is the pre-instruction-tuning checkpoint used in the fine-tuning drift test. Phi-2 (2.7B parameters) (8) is used in the scale-validation run. GPT-2 (125M parameters) is used in the cross-model test.

The embedding model is sentence-transformers/all-MiniLM-L6-v2 (9). The embedding model operates independently of the language models under evaluation. No access to the language model’s hidden states, attention weights, or token probability distributions is required by the protocol. Only the generated text outputs are embedded.

3.3 Test Protocol

Each test follows a three-stage protocol. The preflight stage constructs a plan file that specifies the model, the prompt suite, the sampling parameters, the generation parameters, and the random seeds. The plan file is SHA256 hashed before execution. The prompt suite is embedded directly in the test script and SHA256 hashed as a separate artifact.

The dry-run stage verifies the plan file’s integrity against the hash recorded at preflight and checks that the specified model and embedding model are cached and resolvable.

The execute stage runs real inference on the specified language model, produces the number of generations as specified in the plan file, computes the embeddings using the specified embedding model, computes I , ρ , S , and Φ , and writes the results to disk. A PRE audit record is written before inference begins, containing the plan hash, the prompt hash, the generation parameters, and an environment fingerprint. A POST audit record is written after inference completes, containing the output file hashes and the computed metric values.

No synthetic or fabricated generations are used at any stage. All reported values are derived from real inference on the specified models.

3.4 Hardening Levels

Tests in this paper are reported at three hardening levels. Full hardening includes embedding revision pinning, per-output SHA256 hashes, and a manifest file with SHA256 hashes of all output files. Partial hardening includes embedding revision pinning and per-output hashes but no manifest. Unhardened tests do not pin the embedding revision and do not record per-output hashes.

Tests 3, 6, 7, and 8 are fully hardened. Test 4 is partially hardened. Tests 2 and 5 are not hardened and are carried as supporting evidence. Test 1 is an unhardened protocol validation used to confirm baseline reproducibility rather than a drift detection result.

3.5 Scope and Boundary Conditions

The protocol evaluates Φ under controlled configuration comparisons. Each test fixes a baseline configuration and a comparison configuration, runs both configurations on the same prompt suite with the same embedding model, and reports the resulting Φ values and their difference. This paper evaluates controlled pairwise configuration comparisons rather than unconstrained online deployment monitoring. The protocol does not produce a pass or fail classification; Φ is reported as a continuous measurement and the interpretation of a given Φ value is left to the reader.

Results reported in this paper are bounded to the specific models, prompt suites, configurations, and embedding model used. Generalization beyond these specific choices is not established.

4 Baseline Validation

Test 1 of the eight-test validation suite is a baseline reproducibility check. If the metric is computed on a model compared against itself under identical seeds and identical generation parameters, the expected value of I is 1.0, the expected value of S is 0.0 in

this deterministic self-comparison, and the expected value of Φ is therefore 1.0 in the single-snapshot evaluation. Deviation from these expected values would indicate that the protocol is not implemented correctly or that the model is not behaving deterministically under the specified seeds.

TinyLlama 1.1B-Chat-v1.0 was run against itself under identical seeds and identical generation parameters. The measured values were $I = 1.0$, $S = 0.0$, and $\Phi = 1.0$. Baseline reproducibility confirms that the protocol does not produce spurious drift signals under deterministic conditions. The drift detection tests reported in the following section evaluate whether the protocol produces measurable signals when drift conditions are introduced.

5 Core Findings

5.1 Quality Drift Under Sampling Temperature

The first drift detection test evaluates whether Φ detects quality degradation induced by an increase in sampling temperature. The baseline configuration uses temperature 0.7, a value typical of instruction-tuned generation. The comparison configuration uses temperature 1.5, a value associated with markedly stochastic output. Both configurations run on TinyLlama 1.1B-Chat-v1.0 against the same prompt suite with matched sampling parameters other than temperature. Because Test 2 is a single-snapshot comparison rather than a windowed evaluation, $\rho = 1.0$ by construction and Φ reduces to $I - \alpha \times S$.

The measured values are shown in Table 1. The baseline configuration produced $I = 0.991$, $S = 0.274$, and $\Phi = 0.964$. The comparison configuration produced $I = 0.717$, $S = 0.348$, and $\Phi = 0.682$. The absolute difference in Φ between the two configurations was 0.282. The drop in Φ is driven primarily by the 0.274 decrease in I ; the 0.074 increase in S contributes only 0.0074 to the change in Φ after multiplication by $\alpha = 0.1$.

Configuration	I	S	Φ
Baseline (temperature 0.7)	0.991	0.274	0.964
Comparison (temperature 1.5)	0.717	0.348	0.682
Absolute difference	0.274	0.074	0.282

Table 1: Test 2 measured values. Quality drift on TinyLlama 1.1B-Chat-v1.0 between temperature 0.7 and temperature 1.5 configurations.

Test 2 is not hardened. Embedding revision is not pinned, per-output hashes are not recorded, and no manifest file is produced. The result is carried as supporting evidence for the broader claim that Φ detects drift under configuration changes. The specific magnitude of 0.282 should be interpreted within the scope of the tested prompt suite, the tested temperature pair, and the tested model.

5.2 Safety-Regime Behavioral Drift

The second test evaluates whether Φ detects behavioral drift on safety-sensitive prompts and whether the detected drift co-tracks with changes in refusal behavior. The prompt suite for this test was constructed to include twenty prompts touching on safety-sensitive

topics. The baseline configuration uses temperature 0.2, a value associated with more deterministic generation. The comparison configuration uses temperature 1.0, a value associated with standard instruction-tuned generation. Both configurations run on TinyLlama 1.1B-Chat-v1.0 with five samples per prompt, yielding 200 generations total across both configurations. As in Test 2, the comparison is single-snapshot and $\rho = 1.0$ by construction.

Refusal rates were recorded from the generated outputs.

The measured values are shown in Table 2. The identity similarity between the two configurations was $I_{\text{drift}} = 0.939$. The baseline configuration produced $\Phi_{\text{low}} = 0.985$ and a refusal rate of 14%. The comparison configuration produced $\Phi_{\text{high}} = 0.912$ and a refusal rate of 3%. The absolute difference in Φ between configurations was 0.072. The refusal rate fell by 11 percentage points across the same configuration change.

Configuration	Φ	Refusal rate
Baseline (temperature 0.2)	0.985	14%
Comparison (temperature 1.0)	0.912	3%
Absolute difference	0.072	11 pp

Table 2: Test 3 measured values. Safety-regime drift on TinyLlama 1.1B-Chat-v1.0 across a twenty-prompt safety-sensitive suite at five samples per prompt, yielding 200 generations total across both configurations. The I_{drift} value between configurations was 0.939.

A smoke run at ten prompts and two samples per prompt was conducted before the full run. The smoke run produced $\Phi_{\text{low}} = 0.985$ and $\Phi_{\text{high}} = 0.870$, yielding a Φ drop of 0.115. The direction of the drift signal was consistent between the smoke run and the full run, though the magnitude differed as expected with the smaller sample size.

Test 3 is fully hardened. The embedding revision is pinned to a specific commit, per-output hashes are recorded, and a manifest SHA256 is produced.

The co-tracking between Φ and refusal rate is the key observation from this test. Φ drops and refusal rate drops in the same configuration change, on the same prompt suite, under matched sampling parameters other than temperature. This co-tracking is consistent with the interpretation that Φ responds to a safety-relevant behavioral signal rather than only to generic output variation. The Φ change is bounded to the tested prompt suite and the tested temperature pair; whether Φ co-tracks with refusal rate on out-of-distribution safety categories or on adversarial prompt suites is not established by this test.

5.3 Scale Validation on Phi-2

The third test evaluates whether the Test 3 protocol reproduces on a model with approximately $2.5\times$ the parameter count of TinyLlama. The target model for this scale validation is microsoft/phi-2 with 2.7 billion parameters. The protocol is the same as Test 3 other than a reduced sample count due to the larger model’s generation time: ten prompts at three samples per prompt, yielding 60 generations total across both configurations. The baseline and comparison temperatures are the same as Test 3: 0.2 and 1.0. As in Tests 2 and 3, the comparison is single-snapshot and $\rho = 1.0$ by construction.

The measured values are shown in Table 3. The identity similarity between the two configurations was $I_{\text{drift}} = 0.966$. The baseline configuration produced $\Phi_{\text{low}} = 0.992$ and a refusal rate of 73%. The comparison configuration produced $\Phi_{\text{high}} = 0.948$ and a refusal rate of 53%. The absolute difference in Φ between configurations was 0.044. The refusal rate fell by 20 percentage points across the same configuration change.

Configuration	Φ	Refusal rate
Baseline (temperature 0.2)	0.992	73%
Comparison (temperature 1.0)	0.948	53%
Absolute difference	0.044	20 pp

Table 3: Scale validation on Phi-2 (2.7 billion parameters). Same protocol as Test 3 with reduced sample count. The I_{drift} value between configurations was 0.966.

The Φ drop on Phi-2 is smaller in magnitude than the Φ drop on TinyLlama under the same configuration change. The refusal rates on Phi-2 are higher than on TinyLlama at both temperatures, consistent with Phi-2 producing more refusals on the tested safety-sensitive prompt suite under the same conditions. The smaller Φ drop on Phi-2 is consistent with more stable behavior on this tested model under the tested perturbation. Broader scale trends are not established from this single comparison.

The scale validation confirms that the same formula and the same output-only methodology transfer from TinyLlama to Phi-2 without retraining and without recalibration of α . The Phi-2 run used a reduced sample count for runtime reasons, so the result should be interpreted as a scale-validation extension of Test 3 rather than as a perfectly matched replication. Whether the methodology continues to reproduce on models in the 7B, 13B, or 70B parameter range is not established by this test.

5.4 Temporal Coherence

The fourth test evaluates the temporal coherence component ρ under steady-state conditions. The design runs a fixed prompt suite across six sequential evaluation windows on TinyLlama 1.1B-Chat-v1.0, computes a mean embedding $\bar{e}_w$ per window, and computes ρ as the mean of adjacent-window cosine similarities as defined in the Methods section.

The measured values are as follows. The global mean of ρ across the six windows was 0.989. The global mean of Φ across the six windows was 0.949. The correlation between window index and S was -0.57 ; given the six-window design, this value is reported descriptively and is not interpreted as evidence of temporal drift.

Quantity	Value
ρ (global mean, six windows)	0.989
Φ (global mean, six windows)	0.949
$\text{corr}(\text{window index}, S)$	-0.57

Table 4: Test 4 measured values. Temporal coherence on TinyLlama 1.1B-Chat-v1.0 across six sequential evaluation windows.

Test 4 is partially hardened. The embedding revision is pinned and per-output hashes are recorded, but no manifest file is produced.

Test 4 is a steady-state validation of ρ . It is not a drift detection result. The test confirms that ρ does not drift spontaneously in the absence of an imposed perturbation, which is a prerequisite for interpreting ρ drops in the other tests as meaningful drift signals rather than as baseline variance. The six-window design is a minimal baseline; whether ρ remains steady across longer observation windows or under load conditions is not established by this test.

5.5 Temperature Sensitivity

The fifth test evaluates whether Φ tracks sampling temperature continuously across a wide range. The design evaluates TinyLlama 1.1B-Chat-v1.0 at eight temperature values: 0.3, 0.5, 0.7, 0.9, 1.1, 1.3, 1.5, and 1.7. For each temperature, I , S , and Φ are computed under the same prompt suite and matched sampling parameters other than temperature. Pearson correlations are computed between temperature and each component across the eight points. As in Tests 2, 3, and 3+, the comparison is single-snapshot at each temperature and $\rho = 1.0$ by construction within each condition.

The measured correlations are shown in Table 5. The correlation between temperature and S was $+0.93$, consistent with rising output variation at higher temperatures. The correlation between temperature and I was -0.96 , consistent with falling identity preservation at higher temperatures. The correlation between temperature and Φ was -0.97 .

Pearson correlation	Value
$\text{corr}(\text{temperature}, S)$	$+0.93$
$\text{corr}(\text{temperature}, I)$	-0.96
$\text{corr}(\text{temperature}, \Phi)$	-0.97

Table 5: Test 5 measured values. Pearson correlations between sampling temperature and the three metric components, computed across eight temperature points from 0.3 to 1.7 on TinyLlama 1.1B-Chat-v1.0.

Test 5 is not hardened. Embedding revision is not pinned and per-output hashes are not recorded.

The three correlations are consistent with each other and with the expected effect of temperature on sampling behavior. Higher temperature produces more output variation, which raises S . Higher temperature also produces outputs that are further from the low-temperature baseline, which lowers I . Because $\Phi = I - \alpha \times S$ in this single-snapshot evaluation, both effects push Φ downward, and the observed correlation of -0.97 reflects the combined contribution. The result is bounded to the tested temperature range, the tested model, and the tested prompt suite; whether the same continuous relationship holds on larger models or on out-of-distribution prompt suites is not established by this test.

5.6 Cross-Model Discrimination Under Matched Conditions

The sixth test evaluates whether Φ distinguishes between two tested language models under matched conditions. The two models compared are TinyLlama 1.1B-Chat-v1.0 and GPT-2 (125M parameters). Both models run on the same prompt suite under the same

temperature, the same sampling parameters, and the same embedding model. TinyLlama is treated as the reference configuration in this comparison, so its identity value is 1.000 by definition. As in the prior drift detection tests, the comparison is single-snapshot and $\rho = 1.0$ by construction for each model.

The measured values are shown in Table 6. TinyLlama produced $I = 1.000$, $S = 0.271$, and $\Phi = 0.973$. GPT-2 produced $I = 0.225$, $S = 0.270$, and $\Phi = 0.198$. The absolute difference in Φ between the two models was 0.775. The S values were nearly identical between the two models (0.271 versus 0.270), so the Φ difference is driven almost entirely by the I component.

Model	I	S	Φ
TinyLlama 1.1B-Chat-v1.0 (reference)	1.000	0.271	0.973
GPT-2 (125M)	0.225	0.270	0.198
Absolute difference	0.775	0.001	0.775

Table 6: Test 6 measured values. Cross-model comparison between TinyLlama 1.1B-Chat-v1.0 and GPT-2 under matched conditions.

Test 6 is fully hardened. All model revisions are pinned, per-output hashes are recorded, and a manifest SHA256 is produced.

The Φ gap of 0.775 between the two models is substantially larger than the within-model drift magnitudes observed in Tests 2, 3, 7, and 8. The two models differ along multiple dimensions, including architecture family, parameter count, and training data, and the observed Φ difference reflects their combined differences rather than a single factor. The claim supported by this test is that Φ distinguishes between the two tested models under matched conditions, not that Φ separates arbitrary architecture pairs in general.

5.7 Fine-Tuning Drift

The seventh test evaluates whether Φ detects behavioral drift between a base model checkpoint and its instruction-tuned derivative. The two checkpoints compared are TinyLlama 1.1B-intermediate-step-1431k-3T, the pre-instruction-tuning checkpoint, and TinyLlama 1.1B-Chat-v1.0, the instruction-tuned derivative. Both checkpoints run on the same twenty-prompt suite at five samples per prompt, yielding 100 generations per checkpoint and 200 generations total across both checkpoints. The temperature is fixed at 0.7 in both conditions. The sampling seeds are matched between conditions. As in the prior drift detection tests, the comparison is single-snapshot and $\rho = 1.0$ by construction for each checkpoint.

The measured values are shown in Table 7. The base checkpoint produced $\Phi_{\text{base}} = 0.981$ and a refusal rate of 4%. The instruction-tuned checkpoint produced $\Phi_{\text{ft}} = 0.727$ and a refusal rate of 1%. The I_{drift} between the two checkpoints was 0.743, corresponding to a 26% identity shift between the two. The absolute difference in Φ between the two checkpoints was 0.254. The refusal rate fell by 3 percentage points across the checkpoint change.

Checkpoint	Φ	Refusal rate
TinyLlama 1.1B-intermediate-step-1431k-3T (base)	0.981	4%
TinyLlama 1.1B-Chat-v1.0 (instruction-tuned)	0.727	1%
Absolute difference	0.254	3 pp

Table 7: Test 7 measured values. Fine-tuning drift between the base TinyLlama checkpoint and the instruction-tuned TinyLlama Chat checkpoint, evaluated under matched conditions on a twenty-prompt suite at five samples per prompt. The I_{drift} value between checkpoints was 0.743.

Test 7 is fully hardened. Canonical prompts are enforced, plan and prompt and system hashes are verified, per-output hashes are recorded, and a manifest SHA256 is produced.

The Φ drop of 0.254 between the base and instruction-tuned checkpoints is detected from generated outputs alone, without access to the training process, training data, or training-time logs that produced the fine-tuning. The 26% identity shift is a behavioral measurement rather than a capability judgment. It does not distinguish between fine-tuning that produced a capability gain and fine-tuning that produced a capability loss; it establishes only that the two checkpoints produce different output distributions on the tested prompt suite. Whether Φ detects fine-tuning drift across model families other than TinyLlama, or across fine-tuning regimes other than instruction tuning, is not established by this test.

5.8 Adversarial Injection Drift

The eighth test evaluates whether Φ detects behavioral drift induced by an adversarial prompt injection wrapper. The design runs TinyLlama 1.1B-Chat-v1.0 on a twenty-prompt suite in two conditions. In the normal condition, prompts are presented to the model as written. In the injection condition, a standardized injection wrapper is prepended to each prompt. Both conditions run at five samples per prompt, yielding 100 generations per condition and 200 generations total across both conditions. The temperature is fixed at 0.7 in both conditions. Sampling seeds are matched between conditions to reduce sampling variance as an alternative explanation for the observed drift. As in the prior drift detection tests, the comparison is single-snapshot and $\rho = 1.0$ by construction for each condition.

Leak rates were recorded from the generated outputs as an auxiliary measurement.

The measured values are shown in Table 8. The normal condition produced $\Phi_{\text{normal}} = 0.971$, a refusal rate of 6%, and a leak rate of 12%. The injection condition produced $\Phi_{\text{injection}} = 0.763$, a refusal rate of 4%, and a leak rate of 6%. The I_{drift} between the two conditions was 0.793, corresponding to a 21% identity shift. The absolute difference in Φ between conditions was 0.207.

Condition	Φ	Refusal rate	Leak rate
Normal	0.971	6%	12%
Injection	0.763	4%	6%
Absolute difference	0.207	2 pp	6 pp

Table 8: Test 8 measured values. Adversarial injection drift on TinyLlama 1.1B-Chat-v1.0 across a twenty-prompt suite at five samples per prompt with matched sampling seeds between conditions. The I_{drift} value between conditions was 0.793.

Test 8 is fully hardened. Canonical prompts are enforced, plan and prompt and system and injection hashes are verified, per-output hashes are recorded, a manifest SHA256 is produced, `trust_remote_code` is locked, the embedder fingerprint is recorded, and mean vector hashes are logged for drift re-derivation.

In this test, the evidence for adversarial drift is the measured Φ and I_{drift} change between conditions, not an increase in any single auxiliary rate such as refusal or leak rate. The auxiliary leak-rate measurement did not increase under the tested wrapper. This test measures behavioral drift under an injection wrapper, not the success rate of the wrapper as a harmful-output extraction mechanism.

The Φ drop of 0.207 under the injection wrapper is detected from generated outputs alone. The detection does not require access to input embeddings, token probability distributions, hidden states, attention weights, or any other model internal. The single injection-wrapper design tested here is one point in a much larger space of adversarial attack patterns; whether Φ detects other adversarial patterns, such as multi-turn jail-breaks, encoded payloads, or persona manipulation, is not established by this test. The specific magnitude of 0.207 should not be treated as a universal detection threshold.

6 Discussion

6.1 Returning to the Four Questions

The paper opened with four questions. The results support the following answers, each bounded to the tested settings.

The first question asked whether Φ detects quality degradation induced by sampling temperature on a small instruction-tuned model. Test 2 produced a Φ drop of 0.282 between temperature 0.7 and temperature 1.5 configurations on TinyLlama. The drop was driven primarily by the I component, with the S component contributing only 0.0074 to the change in Φ after multiplication by $\alpha = 0.1$.

The second question asked whether the same metric detects safety-regime behavioral drift on safety-sensitive prompts and whether the detected drift co-tracks with refusal rate changes. Test 3 produced a Φ drop of 0.072 between temperature 0.2 and temperature 1.0 on a twenty-prompt safety-sensitive suite, accompanied by a refusal rate drop from 14% to 3%. The co-occurrence of the two signals under the same configuration change is consistent with the interpretation that Φ tracks a behavioral shift visible on a safety-relevant prompt suite, rather than only generic output variation.

The third question asked whether the metric reproduces on a multi-billion-parameter model with the same formula and methodology. The scale-validation run on Phi-2 produced a Φ drop of 0.044 under the same temperature pair and the same prompt suite as

Test 3, with a reduced sample count for runtime reasons. The Φ drop on Phi-2 is smaller than the Φ drop on TinyLlama. The smaller drop is consistent with more stable behavior on this tested Phi-2 model under the tested perturbation. Broader scale trends are not established from this single comparison.

The fourth question asked whether the metric detects behavioral shifts between a base checkpoint and an instruction-tuned checkpoint, and behavioral shifts induced by adversarial prompt injection with matched sampling seeds. Test 7 produced a Φ drop of 0.254 between the base TinyLlama checkpoint and the instruction-tuned TinyLlama Chat checkpoint. Test 8 produced a Φ drop of 0.207 under an adversarial injection wrapper with matched sampling seeds. Both drops are detected from generated outputs alone, without access to model internals.

6.2 The Role of α and the Weighting of S

The coupling constant α is fixed at 0.1 across all tests. The effect of this choice is to weight the S component one order of magnitude less than the I component in the Φ formula. The S contribution to the Φ change in Test 2 was 0.0074 out of a total change of 0.282, which is approximately 2.6% of the observed drop. In Tests 3, 3+, 6, 7, and 8, the S values between conditions were close enough that their contribution to the Φ differences was correspondingly small. The dominant contributor to the observed Φ drops across all tested conditions was the I component.

This distribution of contribution reflects the choice of α and does not imply that S is uninformative. In Test 5, the correlation between temperature and S was +0.93, which tracks sampling variation closely and contributes meaningfully to the continuous relationship between temperature and Φ . In configurations where I changes little but S changes substantially, the S contribution would be more visible in the Φ signal.

6.3 Black-Box Operation as a Deployment Property

The protocol operates without access to hidden states, attention weights, or token probability distributions. Only the generated text outputs and an external embedding model are required. This property has practical consequences for deployment scenarios in which the monitored language model is accessed through a text-generation API and the model’s internal representations are not exposed to the monitoring system. The method can be applied in such scenarios using only generated text and an external embedding model.

The black-box property also has a methodological consequence. The protocol does not depend on access to the monitored model’s tokenizer internals, hidden states, attention weights, or token probability distributions. Because the same external embedding model can be held fixed across comparisons, outputs from different language models can be compared on the same prompt suite under a common representation, as in Test 6’s cross-model comparison. This supports comparison of Φ values from different monitored models on the same prompt suite under the same embedding model.

6.4 The Co-Tracking Between Φ and Refusal Rate

Test 3’s co-occurrence of a Φ drop and a refusal rate drop is the paper’s strongest single observation. The refusal rate is a direct behavioral measurement on the safety dimension, and its decline from 14% to 3% across the same configuration change that produced the

Φ drop indicates that Φ tracks a behavioral shift visible in the safety-sensitive dimension of the output distribution.

The co-tracking observation is strengthened by the scale-validation run on Phi-2, where the refusal rate also dropped substantially (73% to 53%, a 20 percentage point change) accompanied by a Φ drop. The direction of the co-tracking is consistent across the two models, though the magnitudes differ: the Φ change is smaller on Phi-2 (0.044 versus 0.072), while the refusal-rate change is larger on Phi-2 (20 percentage points versus 11 percentage points). The higher absolute refusal rates on Phi-2 at both temperatures are consistent with Phi-2’s more conservative safety behavior on the tested prompt suite.

The co-tracking is bounded to the tested prompt suite and the tested temperature pair. Whether Φ co-tracks with refusal rate on other safety categories, on adversarial prompt suites, or on production deployment traces is not established by the evidence reported here.

7 Limitations

The evidence reported in this paper has several limitations that bound the interpretation of the results.

The primary test bed across seven of the eight tests is TinyLlama 1.1B-Chat-v1.0. Whether the observed Φ behavior generalizes to other instruction-tuned models at the 1B parameter scale is not established by this evidence.

The scale-validation run uses a single model, microsoft/phi-2, at 2.7B parameters. The scale-validation run used a reduced sample count for runtime reasons, so the Phi-2 result should be interpreted as a scale-validation extension of Test 3 rather than as a perfectly matched replication. Whether the method continues to reproduce on models in the 7B, 13B, or 70B parameter range is not established. The smaller Φ drop on Phi-2 under the same perturbation indicates that detection sensitivity depends on the specific model and that Φ drop magnitudes are not directly comparable across models without calibration.

The safety-sensitive prompt suite used in Tests 3 and 3+ contains twenty prompts. Whether Φ co-tracks with refusal rate on out-of-distribution safety categories, on a larger safety suite, or on adversarial prompts designed to evade detection is not established.

The adversarial injection test uses a single injection-wrapper design. The space of adversarial attack patterns includes multi-turn jailbreaks, encoded payloads, persona manipulation, system-prompt extraction attempts, and indirect injection through retrieved content. Whether Φ detects these other attack patterns at comparable magnitudes is not established.

The fine-tuning drift test compares two checkpoints within the TinyLlama family. Whether Φ detects fine-tuning drift across model families, across fine-tuning regimes other than instruction tuning, or between non-adjacent checkpoints at comparable magnitudes is not established.

Tests 2 and 5 are not hardened. Test 1 is reported as a protocol validation rather than as a drift finding. The Test 3 smoke run is reported as supporting evidence rather than as a primary finding. Test 4 is partially hardened. The paper’s strongest claims rest on the fully hardened Tests 3, 6, 7, and 8.

The embedding model is fixed at sentence-transformers/all-MiniLM-L6-v2 across all tests. Whether the observed Φ behavior persists under different embedding models, dif-

ferent embedding dimensions, or different embedding training domains is not established.

The paper does not establish a threshold for production use. Φ is reported as a continuous measurement, and the interpretation of a given Φ value depends on the specific monitored model, the specific embedding model, the specific prompt suite, and the deployment context. No detection threshold from this paper should be applied to a different monitoring context without calibration.

8 Conclusion

This paper evaluated whether a training-free stability metric computed from generated text outputs alone detects behavioral drift in large language models across four distinct conditions: quality drift under sampling temperature, safety-regime drift on safety-sensitive prompts, fine-tuning drift between a base and instruction-tuned checkpoint, and adversarial drift under prompt injection. The metric is defined as $\Phi = I \times \rho - \alpha \times S$ with the coupling constant fixed at $\alpha = 0.1$.

The reported findings are as follows.

First, Φ detected quality degradation under a temperature increase from 0.7 to 1.5 on TinyLlama 1.1B-Chat-v1.0, with a drop of 0.282.

Second, Φ detected safety-regime behavioral drift on a twenty-prompt safety-sensitive suite under a temperature change from 0.2 to 1.0, with a drop of 0.072 accompanied by a refusal rate drop from 14% to 3%. The co-occurrence of the two signals is consistent with the interpretation that Φ tracks a behavioral shift visible on a safety-relevant prompt suite, rather than only generic output variation.

Third, the same formula and the same output-only evaluation framework, with a reduced sample count for runtime reasons, reproduced on Phi-2 (2.7 billion parameters) with a Φ drop of 0.044 under the same temperature pair and prompt suite, supporting the claim that the protocol transfers from a 1.1B model to a 2.7B model without retraining.

Fourth, Φ detected fine-tuning drift between the base TinyLlama checkpoint and the instruction-tuned Chat checkpoint with a drop of 0.254, and adversarial drift under a prompt injection wrapper with matched sampling seeds with a drop of 0.207. Both drops were detected from generated outputs alone without access to model internals.

Fifth, the protocol operated in black-box settings across all eight tests, using only an external embedding model and generated text outputs. No access to hidden states, attention weights, or token probability distributions was required to produce any of the four drift-detection findings above.

The results are bounded to the specific models, prompt suites, configurations, and embedding model used. Generalization to larger models beyond the 2.7B parameter scale, to out-of-distribution safety categories, to other fine-tuning regimes, or to adversarial attack patterns beyond the tested injection wrapper is not established by the evidence reported here.

Code Availability

The reference implementation of the stability metric, the validation test scripts, the prompt suites, the plan files, and repository documentation describing the audit structure are available in the public repository at <https://github.com/Wise314/llm-phi-stability>.

The core metric computation is implemented in `core/phi_llm.py` and accepts generated text outputs, a prompt suite specification, and generation parameters. The computation does not require access to the monitored language model’s hidden states, attention weights, or token probability distributions.

Patent Disclosures

The methods described in this paper are the subject of U.S. Provisional Patent Application No. 63/973,673 (filed February 2, 2026). No license to implement or commercialize the described methods is granted by this publication. All rights reserved.

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